

MHDRM Cross-Site Readiness Assessment

Prototype screening report for multimodal federated learning

Assessment Context

Project name	AI-Enabled Patient Monitoring
MMFL use case	Remote patient monitoring application which identifies falls and sends an alert to medical staff.
Target outcome or label	Binary event-level outcome: fall / no fall.
Participating sites	Nursing home 1 Nursing home 2 Nursing home 3 Commercial patient monitoring company University research department
Target population	Nursing home residents who are at risk of falls.
Required modalities	Wearable device data, Other

Assessment Summary

Total score	Readiness	Classification
9/14	64.3%	Partial cross-site compatibility

Pillar Results

Category	MHDRM pillar	Response	Score	Flag
Data Characteristics	Modality Availability	1 — Partially demonstrated	1	
Data Characteristics	Data Quality and Completeness	2 — Demonstrated	2	
Data Characteristics	Population Representativeness	1 — Partially demonstrated	1	
Integration Readiness	Standardization and Interoperability	2 — Demonstrated	2	
Integration Readiness	Cross-Modal Alignment	1 — Partially demonstrated	1	
Organizational Readiness	Privacy and Security by Modality	1 — Partially demonstrated	1	
Organizational Readiness	Provenance and Governance	1 — Partially demonstrated	1	

Evidence and Rationale

MHDRM pillar	Evidence reviewed and scoring rationale
Modality Availability	An audit of covered modalities is available from every site, with documented data quality challenges identified.
Data Quality and Completeness	Site audits documented the frequency and duration of missing or delayed observations caused by intermittent network connectivity.
Population Representativeness	Target population is established and documented, but relevant subgroups by age, mobility, cognitive status, and assistive device use have not yet been assessed.

MHDRM pillar	Evidence reviewed and scoring rationale
Standardization and Interoperability	All sites use the same audio, video, and wearable devices with standardized configuration and data formats.
Cross-Modal Alignment	Intermittent network issues at the nursing homes sometimes interrupt device data flows, preventing records from being aligned within the specified detection window.
Privacy and Security by Modality	Privacy controls exist and are documented, but deidentification processes for audio and video data differs across sites.
Provenance and Governance	Data collaboration agreements exist, but responsibility for ongoing auditing is not established.

Prototype limitation: This report provides a preliminary preimplementation screening of cross-site data compatibility. The scores and thresholds have not been empirically validated and do not certify that an MMFL initiative will be successful.